

BOWLING GREEN 21 NNE, KY, USA

WMO: 724130

Lat: 37.2504N

Lon: 86.2325W

Elev: 241

StdP: 98.47

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63849

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.9	(c) -9.3	(d) -18.1	(e) 0.8	(f) -9.1	(g) -15.4	(h) 1.0	(i) -8.0	(j) 4.3	(k) 12.5	(l) 3.9	(m) 12.0	(n) 0.5	(o) N/A	(p) 0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.5	(c) 32.8	(d) 23.8	(e) 31.6	(f) 23.7	(g) 30.6	(h) 23.1	(i) 25.4	(j) 30.4	(k) 24.8	(l) 29.7	(m) 24.1	(n) 28.8	(o) 1.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.4	28.4	23.2	18.5	27.6	22.7	17.9	26.9	79.0	30.4	76.3	29.9	73.8	28.8	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
3.7	3.1	2.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-15.1	34.8	3.0	2.3	-17.3	36.5	-19.0	37.9	-20.7	39.2	-22.8	40.9		
			-15.8	26.5	3.2	0.8	-18.1	27.0	-20.0	27.5	-21.8	28.0	-24.1	28.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	2.0	3.9	9.2	15.0	19.4	23.4	24.6	24.5	21.4	15.2	8.9	4.5		
	DBStd	9.49	6.80	6.71	6.28	5.29	4.35	2.63	2.40	2.62	3.66	5.15	5.40	5.81		
	HDD10.0	847	261	191	93	17	1	0	0	0	0	12	83	189		
	HDD18.3	2223	507	404	288	123	44	2	1	1	14	124	286	429		
	CDD10.0	2448	12	20	69	167	291	402	454	449	343	174	49	18		
	CDD18.3	784	0	1	6	23	76	154	196	192	107	28	2	0		
	CDH23.3	6111	0	1	33	167	537	1199	1624	1559	791	191	7	0		
	CDH26.7	2015	0	0	3	22	126	385	591	583	261	43	1	0		
(14) Wind	WSAvg	1.0	1.2	1.3	1.4	1.4	1.1	0.9	0.8	0.7	0.7	0.9	1.0	1.1		
(15) Precipitation	PrecAvg	1288	105	95	128	113	142	109	112	85	88	96	92	121		
	PrecMax	1574	202	172	366	201	310	203	207	231	203	215	165	270		
	PrecMin	992	57	33	60	47	63	53	35	16	26	10	22	47		
	PrecStd	148	33	40	71	42	65	42	40	43	55	56	33	55		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	22.1	26.8	28.9	30.8	32.8	34.4	35.2	33.7	30.3	24.1	20.3		
		MCWB	14.9	15.5	16.9	18.2	22.0	23.5	24.5	25.6	22.3	21.9	15.6	16.0		
	2%	DB	16.7	19.1	23.9	27.0	29.4	31.4	32.3	33.0	31.4	27.7	21.6	17.9		
		MCWB	13.3	14.0	15.4	17.1	20.7	23.3	24.5	24.3	22.4	19.2	14.9	14.1		
	5%	DB	15.1	16.8	21.4	25.4	28.2	30.4	31.3	31.3	29.8	25.5	19.3	15.7		
		MCWB	11.9	12.8	14.0	16.2	20.2	22.9	24.3	23.4	21.9	18.0	13.6	12.8		
	10%	DB	12.8	14.4	19.0	23.5	26.7	29.1	30.1	30.0	27.9	23.1	17.1	13.5		
		MCWB	9.3	10.9	12.2	15.7	19.9	22.4	23.8	22.8	21.2	17.6	12.0	10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	17.0	17.8	20.3	23.5	25.5	26.7	26.1	24.6	23.2	18.0	17.4		
		MCDB	18.3	20.5	24.3	26.6	28.7	30.1	31.5	31.9	29.6	28.0	21.7	19.6		
	2%	WB	13.8	15.7	16.5	19.3	22.4	24.7	25.6	25.0	23.7	21.6	16.5	15.3		
		MCDB	15.8	19.0	22.3	24.9	27.7	29.8	30.6	29.7	29.0	25.0	19.5	17.8		
	5%	WB	11.6	14.0	15.2	18.0	21.6	24.0	25.1	24.3	22.8	19.9	14.7	13.5		
		MCDB	13.8	16.1	19.6	22.9	26.7	29.1	30.1	28.9	27.9	23.5	17.7	15.6		
	10%	WB	9.3	11.8	13.7	16.8	20.8	23.2	24.4	23.6	22.0	18.0	12.7	11.3		
		MCDB	11.6	14.4	17.8	21.6	25.6	28.0	29.1	27.9	26.3	22.1	16.0	13.6		
(35) Mean Daily Temperature Range	MDBR		8.6	9.3	10.3	11.0	10.1	9.8	9.5	9.9	10.5	10.7	9.8	8.2		
	5% DB	MCDBR	10.2	10.9	12.3	12.1	11.2	10.9	10.6	11.3	11.8	12.1	11.6	9.8		
		MCWBR	8.6	8.2	7.4	5.9	5.2	5.0	4.6	4.8	4.7	5.6	6.5	7.2		
	5% WB	MCDBR	10.3	10.2	10.9	10.5	9.8	10.0	9.6	9.3	10.0	10.0	9.5	9.2		
		MCWBR	9.1	8.6	6.7	6.2	5.0	5.1	4.7	4.5	4.5	5.7	7.1	7.9		
(40) Clear-Sky Solar Irradiance	taub		0.312	0.321	0.361	0.405	0.443	0.466	0.497	0.510	0.422	0.362	0.330	0.306		
	taud		2.470	2.425	2.369	2.244	2.183	2.149	2.081	2.029	2.260	2.421	2.467	2.493		
	Ebn at Noon		877	912	902	876	845	822	793	772	827	853	847	859		
	Edh at Noon		82	97	113	136	147	152	162	167	125	97	82	76		
(44) All-Sky Solar Radiation	RadAvg		2.04	2.68	3.77	5.00	5.58	6.25	5.95	5.52	4.61	3.50	2.45	1.76		
	RadStd		0.21	0.36	0.36	0.35	0.51	0.36	0.35	0.30	0.41	0.40	0.26	0.17		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)		
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+82	(47)	

Nomenclature: See separate page